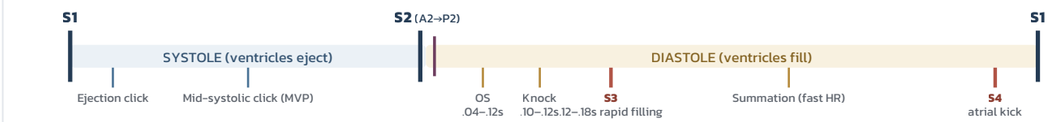
HEART SOUNDS

S1 · S2 · S3 · S4 · Splitting of S2 · Loud P2/A2 · Gallops · Extra sounds — one-page table

Abbrev: LLSB/LUSB = left lower/upper sternal border · ICS = intercostal space · LLD = left lateral decubitus · CMP = cardiomyopathy · PHT = pulmonary hypertension · HF = heart failure

Where each sound falls in one cardiac cycle



S2 splitting at a glance (■ A2 · ■ P2)



Sound	Definition (whole)	Word-by-word	When / phase heard	Pathophysiology (step by step)	Inspiration vs expiration	Diseases / conditions	Where & how to listen	Key pearl
A · NORMAL HEART SOUNDS								
S1 "lub"	First heart sound = closure of the **AV valves** (mitral M1 → tricuspid T1) at the start of ventricular systole.	**S** = sound · **1** = first in the cycle · M1 = mitral, T1 = tricuspid component.	**Start of systole** (isovolumetric contraction). Just after QRS; with carotid upstroke.	❶ Ventricle depolarizes (QRS) → contracts ❷ LV pressure rises > LA pressure ❸ Mitral (then tricuspid) leaflets snap shut ❹ Sudden deceleration of blood → vibration of valves/walls = S1	Little change. M1–T1 split (rarely heard, LLSB) may widen slightly on inspiration.	**Loud:** short PR, early MS, tachycardia/hyperdynamic (fever, anemia, thyrotoxicosis). **Soft:** long PR (1° AV block), MR, calcified MS, LV failure, obesity/COPD/effusion. **Variable:** AF, complete heart block.	**Apex** (M1), LLSB (T1). **Diaphragm**.	Loudest at apex (S1 > S2 there). Time it with the carotid pulse to identify systole.
S2 "dub"	Second heart sound = closure of the **semilunar valves** (aortic A2 → pulmonic P2) at the end of systole.	**S** = sound · **2** = second · A2 = aortic, P2 = pulmonic closure.	**End of systole / start of diastole** (isovolumetric relaxation). ~ end of T wave.	❶ Ejection ends; ventricles relax ❷ Ventricular pressure falls < aortic & pulmonary artery pressure ❸ Backflow snaps aortic cusps shut (A2), then pulmonic (P2) ❹ P2 later: low-resistance, compliant pulmonary bed → longer 'hangout interval'	Normally **splits on inspiration**, single on expiration (see C).	Normal. Abnormal forms → C & D. **Single S2:** elderly (P2 inaudible), severe AS, pulmonary atresia, TGA, truncus, emphysema.	**Base**: 2nd RICS (A2), 2nd–3rd LICS (P2 & split). **Diaphragm**.	A2 is normally louder everywhere; P2 heard only at pulmonic area. S2 > S1 at the base.
B · EXTRA DIASTOLIC SOUNDS & GALLOPS								
S3 "ventricular gallop"	Low-pitched **early-diastolic** sound from rapid passive filling of a volume-loaded/dilated ventricle.	**S3** = third sound · **Ventricular** = arises in ventricle · **Protodiastolic** = early (proto) diastole.	**Early diastole**, 0.12–0.18 s after S2 — **rapid ventricular filling** phase.	❶ AV valve opens → atrial blood rushes into ventricle ❷ High volume/flow or dilated, poorly compliant ventricle ❸ Ventricle reaches its elastic limit → inflow decelerates abruptly (chordae tense) ❹ Vibration of ventricular wall = S3	LV S3 ↑ on **expiration**; RV S3 ↑ on **inspiration**.	**Physiologic** (<40 y): children, athletes, pregnancy, high output. **Pathologic:** HFrEF, dilated CMP, severe MR/TR, AR, large VSD/PDA, thyrotoxicosis, anemia.	**Bell**, light pressure, **apex**, **left lateral decubitus**, end-expiration. RV: LLSB.	Cadence "**Ken-tuc-KY**" (S1-S2-S3). Pathologic after 40. **Absent in severe MS** (inflow blocked).
S4 "atrial gallop"	Low-pitched **late-diastolic** sound from atrial contraction into a **stiff** ventricle.	**S4** = fourth sound · **Atrial** = caused by atrial kick · **Presystolic** = just before systole.	**Late diastole**, just before S1 (after P wave) — **atrial systole** phase.	❶ Hypertrophied/ischemic ventricle → ↓ compliance ❷ Atrium contracts forcefully (atrial kick) to fill it ❸ Blood meets non-compliant wall → abrupt wall tensing ❹ Vibration = S4 (needs atrial contraction!)	LV S4 ↑ on **expiration**; RV S4 ↑ on **inspiration**.	HTN heart disease/LVH, **aortic stenosis**, **HCM**, acute MI/ischemia, restrictive CMP, elderly. RV: pulmonary HTN, PS.	**Bell**, **apex**, **left lateral decubitus**. RV: LLSB.	Cadence "**TEN-nes-see**" (S4-S1-S2). Almost always pathologic. **Never in AF** (no atrial kick).
S3 gallop	Triple rhythm **S1-S2-S3** at a fast rate resembling a galloping horse = pathologic S3.	**Gallop** = 3-beat cadence of a horse · S3 = the added early-diastolic beat.	Early diastole (rapid filling).	❶ Failing LV → ↑ end-systolic volume & filling pressure ❷ ↑ LA pressure → forceful early inflow into dilated LV ❸ Abrupt deceleration → loud S3 ❹ Compensatory tachycardia → galloping cadence	LV: louder on **expiration**.	Decompensated **HFrEF**, dilated CMP, acute MI with LV failure, severe MR, high-output HF.	Bell, apex, LLD.	Sign of ↑ filling pressure; poor prognosis; fades with diuresis.
S4 gallop	Triple rhythm **S4-S1-S2** at a fast rate = pathologic S4.	**Gallop** + S4 = the added presystolic beat.	Late diastole (atrial systole).	❶ Stiff/hypertrophied or ischemic LV ❷ Strong atrial kick against high LVEDP ❸ Abrupt wall tensing → loud S4 ❹ Tachycardia → gallop cadence	LV: louder on **expiration**.	HTN crisis/LVH, AS, HCM, **acute ischemia** (during angina/MI), restrictive CMP.	Bell, apex, LLD.	Diastolic (relaxation) dysfunction marker. Absent in AF.
Summation gallop	S3 + S4 fuse into one loud mid-diastolic sound when HR is fast (>~120).	**Summation** = adding together. (At slower HR both = **quadruple rhythm**.)	Mid-diastole.	❶ Tachycardia shortens diastole ❷ Rapid-filling & atrial-kick phases overlap ❸ S3 + S4 superimpose → one loud sound	LV: louder on expiration.	Advanced HF with tachycardia.	Bell, apex.	Slow the HR (e.g., carotid massage) → may separate into S3 & S4.
C · SPLITTING OF S2								
Split S2 (general)	Audible separation of S2 into **A2** and **P2**.	**Split** = divided in two · S2 = second sound.	S2. Normal gap <30 ms (exp) → 40–80 ms (insp).	❶ Anything that **delays P2** (long RV ejection, RBBB, ↑ pulm. capacitance) or **advances A2** (short LV ejection) → wider split ❷ Anything that **delays A2** → narrow/reversed split	Depends on type (below).	Normal · wide · fixed · paradoxical · single.	**2nd–3rd LICS**, diaphragm, supine (sitting may abolish physiologic split).	Always ask 2 questions: **Is it split on expiration? Does it change with breathing?**
Physiologic split	Normal split that **appears on inspiration** and **disappears on expiration**.	**Physiologic** = normal body function (not disease).	S2 during **inspiration**.	❶ Inspiration → ↓ intrathoracic pressure ❷ ↑ venous return to RA/RV → ↑ RV stroke volume → longer RV ejection → **P2 later** ❸ ↑ pulmonary capacitance (longer hangout) → P2 later ❹ Blood pools in lungs → ↓ LV return → shorter LV ejection → **A2 earlier** ❺ Expiration reverses → single S2	Insp: **split** · Exp: **single**.	Normal — esp. children & young adults.	Pulmonic area, supine.	Most obvious in the young; often inaudible in elderly.
Wide split	Split heard on **both** phases, **widening further on inspiration** (respiratory variation preserved).	**Wide** = gap larger than normal.	S2, both phases.	❶ **Delayed P2:** RBBB (late RV activation), pulmonic stenosis/PHT/PE (prolonged RV ejection) ❷ **Early A2:** MR, VSD (LV empties by a second outlet → short LV ejection) ❸ Normal respiratory effect still added on top	Insp: **wider** · Exp: **still split**.	**RBBB**, pulmonic stenosis, acute PE, pulmonary HTN with RV failure, **MR**, **VSD**, WPW (LV pre-excitation).	Pulmonic area.	Key difference from fixed split: it still **varies** with breathing.
Fixed split = Wide fixed split	Wide split that **does not change** with respiration.	**Fixed** = constant gap · **Wide** = larger than normal (fixed split is always wide).	S2, both phases (same width).	❶ ASD: L→R shunt at atrial level → chronic RV volume overload → long RV ejection → **late P2** (wide) ❷ **Insp:** ↑ systemic return to RA but ↑ RA pressure → ↓ L→R shunt ❸ **Exp:** ↓ systemic return but ↑ L→R shunt ❹ Total RV filling stays ~constant (atria share pressure) → gap **fixed**	Insp = Exp (**no change**).	**ASD** (secundum = classic), partial anomalous pulmonary venous return, severe RV failure.	Pulmonic area.	ASD also: pulmonic flow ejection murmur (LUSB) ± tricuspid flow rumble — murmur is **not** from the hole.
Paradoxical split = Reversed split	**A2 delayed after P2**: split heard on **expiration**, disappears on **inspiration**.	**Paradoxical** = opposite of expected · **Reversed** = order becomes P2 → A2.	S2 during **expiration**.	❶ Delayed LV activation or prolonged LV ejection → **A2 after P2** ❷ Insp: P2 moves later → toward A2 → **single** ❸ Exp: P2 moves earlier → away from A2 → **split**	Insp: **single** · Exp: **split**.	**LBBB** (most common), RV pacing, **severe AS**, HOCM, severe HTN, acute ischemia/LV failure, large PDA, WPW (RV pre-excitation).	Pulmonic area.	Think "left side is late": LBBB, AS, LV failure.
Single S2	Only **one** S2 component is heard on **both** inspiration and expiration (no audible split).	**Single** = one · **S2** = second sound (A2 & P2 fused, or one of them absent).	S2, both phases.	❶ **A2 absent:** calcified, immobile aortic valve (severe AS) ❷ **P2 inaudible/absent:** elderly, emphysema/obesity, pulmonary atresia, TOF (tiny stenotic P valve), TGA (P valve posterior, aorta anterior) ❸ **Only one semilunar valve:** truncus arteriosus ❹ **A2 & P2 coincide:** severe PHT/Eisenmenger (narrow split), paradoxical split during inspiration	Insp = Exp (**single in both**).	Elderly (normal), emphysema, severe **AS**, pulmonary atresia, **Tetralogy of Fallot**, **truncus arteriosus**, **TGA**, Eisenmenger/severe PHT.	Pulmonic area, diaphragm, supine.	Older adult: usually benign (P2 too soft). Cyanotic newborn + single S2 → think TOF, truncus, TGA.
D · INTENSITY OF S2 COMPONENTS								
Loud P2	Accentuated **pulmonic** component of S2 (P2 ≥ A2 at pulmonic area, or P2 heard at the apex).	**Loud** = ↑ intensity · **P2** = pulmonic valve closure.	S2, after A2 (end of RV systole).	❶ ↑ Pulmonary artery pressure ❷ ↑ Diastolic pressure gradient across pulmonic valve ❸ Cusps slam shut more forcefully → loud P2	Louder & more separated on **inspiration**.	**Pulmonary HTN**: PAH, COPD/ILD (cor pulmonale), chronic/acute PE, left HF, **mitral stenosis**, L→R shunts/Eisenmenger, OSA, altitude.	2nd LICS (and apex), diaphragm.	May be palpable + RV heave. Normally louder in children/thin chests.
Loud A2	Accentuated **aortic** component of S2.	**Loud** = ↑ intensity · **A2** = aortic valve closure.	S2 (first component).	❶ ↑ Aortic diastolic pressure (HTN) → stronger closing force ❷ Or dilated aortic root sits closer to chest wall → 'ringing/tambour' A2	Little change (both phases).	**Systemic HTN**, aortic root dilation (Marfan, aortitis/syphilis, aneurysm), coarctation, TGA (anterior aorta). **Soft A2:** calcified AS, AR.	2nd RICS, diaphragm.	Soft/absent A2 + ejection murmur = severe (calcified) AS.
E · OTHER RELEVANT EXTRA SOUNDS (not listed but high-yield)								
Opening snap (OS)	High-pitched early-diastolic sound from a **stenotic mitral valve** snapping open.	**Opening** = valve opening · **Snap** = sharp, crisp.	Early diastole, **0.04–0.12 s after A2** (earlier than S3).	❶ Rheumatic commissural fusion → ↑ LA pressure ❷ When LV pressure < LA, domed leaflets open ❸ Abrupt tensing of stiff leaflets → snap ❹ Followed by diastolic rumble	Little change.	**Mitral stenosis** (rheumatic); rarely tricuspid stenosis.	Apex–LLSB, **diaphragm**.	**Shorter A2–OS = more severe MS.** Lost when valve heavily calcified.
Ejection click	High-pitched **early-systolic** sound just after S1.	**Ejection** = at onset of ejection · **Click** = short, sharp.	Early systole.	❶ Stenotic-but-mobile semilunar valve (or dilated root) ❷ Leaflets dome abruptly at opening → click	Aortic: no change. **Pulmonic: softer on inspiration** (only right-sided exception).	Bicuspid aortic valve, congenital AS, pulmonic stenosis, PHT, dilated aorta.	Aortic: apex/base. Pulmonic: LUSB.	Disappears as the valve calcifies (immobile).
Mid-systolic click	Mid–late systolic click of **mitral valve prolapse**.	**Mid-systolic** = middle of systole.	Mid systole.	❶ Redundant leaflet prolapses into LA ❷ Chordae/leaflet suddenly tense → click ± late systolic murmur	No significant change.	**MVP** (myxomatous; Marfan, Ehlers-Danlos).	Apex, diaphragm.	↓ LV volume (standing, Valsalva) → **earlier** click; squat/handgrip → later.
Pericardial knock / Tumor plop	Knock: high-pitched early-diastolic sound in **constrictive pericarditis**. Plop: **atrial myxoma** prolapsing into ventricle.	**Knock** = abrupt stop · **Plop** = tumor dropping.	Early diastole (knock earlier than S3).	❶ Knock: rigid pericardium abruptly halts early filling ❷ Plop: pedunculated tumor falls through AV valve and stops	Knock: ↑ with inspiration. Plop: varies with **position**.	Constrictive pericarditis (TB, post-surgery, radiation) · left atrial myxoma.	LLSB/apex, diaphragm.	Knock + Kussmaul sign. Plop + positional murmur, syncope, emboli.

Respiration rule (Carvallo)
Inspiration ↑ venous return to **right** heart → **right-sided** sounds/murmurs louder. Expiration → **left-sided** louder. Exception: pulmonic ejection click **softer** on inspiration.

Bell vs diaphragm
Bell (light touch) = low pitch: S3, S4, MS rumble. **Diaphragm** (firm) = high pitch: S1, S2, splits, clicks, OS, knock, AR. Firm bell pressure makes S3/S4 fade.

Early-diastolic sound after S2?
Split S2 (≤0.06 s, base, both high) → **OS** (0.04–0.12 s, high, MS) → **Knock** (0.10–0.12 s, high) → **S3** (0.12–0.18 s, **low**, apex, bell).

Sound before S1?
S4 = low pitch, apex, bell, gone in AF. **Split S1** = both high pitch, LLSB, diaphragm. **Ejection click** = after S1, high pitch.

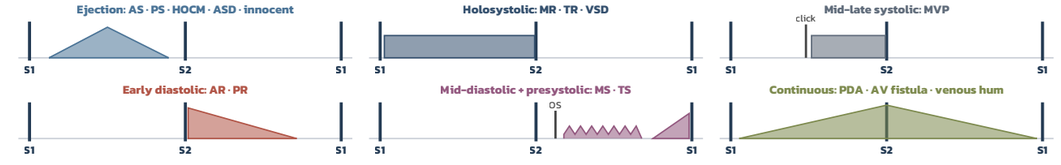
Auscultation areas
Aortic 2nd RICS · **P**ulmonic 2nd LICS · Erb's 3rd LICS · **T**ricuspid 4th LICS/LLSB · **M**itral apex 5th ICS MCL ("All Physicians Take Money").

HEART MURMURS

Systolic · diastolic · continuous · eponymous murmurs · pericardial rub — one-page table

Abbrev: LSB/RSB = left/right sternal border · LLSB/LUSB = lower/upper LSB · ICS = intercostal space · LLD = left lateral decubitus · PHT = pulmonary hypertension · SAM = systolic anterior motion · LVOT = LV outflow tract · CMP = cardiomyopathy

Murmur shapes in the cardiac cycle



Dynamic auscultation (maneuvers)

Maneuver	Effect	Louder	Softer
Inspiration	↑ venous return to right heart	Right-sided (TR, TS, PR, PS)	Left-sided
Valsalva / standing	↓ preload	**HOCM, MVP** (earlier click)	Most others (AS, MR…)
Squat / leg raise	↑ preload (+ afterload)	AS, MR, AR, VSD	HOCM, MVP (later click)
Handgrip	↑ afterload	MR, AR, VSD	AS, HOCM
Amyl nitrite	↓ afterload	AS, HOCM, MS	MR, AR, VSD, Austin Flint

Murmur	Definition (whole)	Word-by-word	When / phase heard	Pathophysiology (step by step)	Inspiration, expiration & maneuvers	Diseases / conditions	Where & how to listen	Key pearl
A · SYSTOLIC EJECTION MURMURS (crescendo–decrescendo / diamond)								
Aortic stenosis AS	Harsh **crescendo–decrescendo** systolic ejection murmur from flow through a **narrowed aortic valve**.	**Aortic** = aortic valve · **Stenosis** (Gk stenos) = narrowing.	**Midsystole** (ejection): starts after S1 (± ejection click), ends before A2.	❶ Calcification/fusion → ↓ valve area ❷ LV must generate high pressure → LV–aortic gradient ❸ High-velocity turbulent jet rises & falls with ejection → diamond murmur ❹ Pressure overload → concentric LVH → S4; severe: late-peaking murmur, soft A2, paradoxical split	Louder on **expiration**. ↓ Valsalva/ standing/handgrip; ↑ squat/leg raise; ↑ beat after PVC.	**Senile calcific** (>65), **bicuspid valve** (<65), rheumatic (with MS). Symptoms: **S**yncope, **A**ngina, **D**yspnea/HF.	**2nd RICS**, diaphragm, sitting forward. Radiates to **carotids**.	Pulsus **parvus et tardus**. Later peak = more severe. Aortic **sclerosis**: same murmur, normal pulse & A2.
Hypertrophic obstructive CMP HOCM	Systolic ejection murmur from **dynamic LV outflow obstruction** by a thick septum ± SAM of mitral leaflet.	**Hypertrophic** = thickened · **Obstructive** = blocks outflow · **Cardio-myo-pathy** = heart-muscle disease.	**Midsystole**; often with late MR.	❶ Sarcomere mutation → asymmetric septal hypertrophy ❷ Narrow LVOT → fast flow drags anterior mitral leaflet forward (Venturi → **SAM**) ❸ Dynamic obstruction + MR → murmur ❹ **Smaller LV cavity = more obstruction**	Little resp change. ↑ **Valsalva/standing**; ↓ squat, handgrip, leg raise.	Familial HCM (AD; β-MHC, MYBPC3); sudden death in young athletes.	**LLSB**/apex. **No** carotid radiation.	With MVP, the only murmur ↑ by Valsalva. S4, bisferiens pulse.
Pulmonic stenosis PS	Systolic ejection murmur from flow through a **narrowed pulmonic valve**.	**Pulmonic** = pulmonary valve · **Stenosis** = narrowing.	**Midsystole**, after an ejection click (valvular).	❶ Fused/dysplastic valve → RV–PA gradient ❷ Turbulent jet → murmur ❸ Prolonged RV ejection → late, soft P2 → wide split ❹ RV pressure overload → RVH, RV heave	↑ on **inspiration** (click ↓ on inspiration).	Congenital, **Noonan**, **Tetralogy of Fallot**, congenital rubella, carcinoid.	**2nd LICS**; radiates to left shoulder/ back.	Longer murmur + later P2 = more severe.
Atrial septal defect ASD (flow murmur)	Systolic ejection murmur from ↑ **flow across a normal pulmonic valve** due to atrial L→R shunt.	**Atrial Septal Defect** = hole in the wall between the atria.	**Midsystole** + wide **fixed** split S2 ± mid-diastolic tricuspid flow rumble.	❶ LA pressure > RA → L→R shunt ❷ RV & pulmonary volume overload ❸ ↑ flow across normal pulmonic valve (relative PS) → murmur ❹ Flow across ASD itself: low gradient → **silent**	Split fixed; murmur mildly ↑ inspiration.	Secundum (most), primum (Down syndrome), sinus venosus; paradoxical emboli; Eisenmenger.	2nd–3rd LICS.	The hole is silent — the key sign is the **fixed split**.
Innocent / flow murmurs Still's · pulmonary flow	Soft systolic ejection murmurs in a **structurally normal heart**.	**Innocent** = harmless · **Functional / flow** = from ↑ flow, not disease.	Early–midsystole, grade ≤2/6.	❶ Normal or ↑ stroke volume (high-output state) ❷ Fast flow through normal outflow tracts ❸ Mild turbulence/vibration of normal structures	Changes with **posture**; louder supine, softer standing.	Children (**Still's** = musical/vibratory, LLSB; pulmonary flow, LUSB), pregnancy, anemia, fever, thyrotoxicosis, athletes.	LLSB / LUSB.	**7 S**: Soft, Systolic, Short, Single S2 normal, Small area, Sweet, Sensitive to position.
B · HOLOSYSTOLIC (PANSYSTOLIC) MURMURS (plateau, S1 → S2)								
Chronic mitral regurgitation MR	**Blowing holosystolic** murmur from backflow **LV → LA** through an incompetent mitral valve.	**Mitral** = bishop's-mitre valve · **Regurgitation** = backward flow.	**Holosystolic**: begins with S1, extends to/through A2 (plateau).	❶ Leaflet/annulus/chordae/papillary dysfunction → poor coaptation ❷ LV > LA pressure from S1 to S2 → leak all systole → plateau murmur ❸ LA & LV volume overload → dilation → S3, AF ❹ ↓ forward flow → HF	↑ **expiration**. ↑ handgrip/squat (afterload); ↓ Valsalva, amyl nitrite.	MVP, rheumatic, ischemic papillary dysfunction, **dilated CMP** (functional), endocarditis, HCM.	**Apex** → radiates to **axilla**; diaphragm, LLD.	Soft S1 + S3 = severe MR. Early A2 → wide split S2.
Tricuspid regurgitation TR	Holosystolic murmur from backflow **RV → RA**.	**Tricuspid** = three-cusped right AV valve · **Regurgitation**.	**Holosystolic**.	❶ RV dilation (mostly from PHT) → annulus stretches ❷ Poor coaptation → RV → RA leak all systole ❸ ↑ RA pressure → giant **v (CV) waves**, pulsatile liver, edema	↑ on **inspiration** (**Carvallo sign**).	Functional (PHT, left HF), **IV-drug endocarditis**, carcinoid, Ebstein anomaly, rheumatic, pacemaker lead.	**LLSB**, radiates to right sternal border.	Look at JVP (CV waves) & liver.
Ventricular septal defect VSD	**Harsh** holosystolic murmur from L→R shunt across the ventricular septum.	**Ventricular Septal Defect** = hole in the wall between the ventricles.	**Holosystolic**.	❶ LV ≫ RV pressure throughout systole ❷ High-velocity jet through defect → harsh murmur + thrill ❸ ↑ pulmonary flow → LA/LV overload (± mitral flow rumble) ❹ Chronic → PHT → **Eisenmenger**: shunt reverses, murmur fades, cyanosis	Little change. ↑ handgrip.	**Most common congenital** defect; Down syndrome, TOF, **post-MI septal rupture** (3–5 days).	**LLSB** (3rd–4th LICS); thrill.	**Smaller hole = louder murmur**.
C · EARLY– & LATE–SYSTOLIC MURMURS								
Mitral valve prolapse MVP	**Mid-systolic click** followed by a **late-systolic** murmur.	**Prolapse** = slipping forward/down (leaflet bulges into LA).	**Mid-to-late systole**, after the click.	❶ Myxomatous degeneration → redundant leaflets, long chordae ❷ Mid-systole: leaflet billows into LA → chordae snap tight = **click** ❸ Coaptation lost → late-systolic MR	No resp change. **Valsalva/standing → earlier click, longer murmur**; squat → later.	Primary myxomatous, **Marfan**, Ehlers-Danlos, ADPKD.	**Apex**, diaphragm.	Most common valve lesion; may progress to MR/ chordal rupture.
Acute mitral regurgitation	Short, **early-systolic decrescendo** murmur of sudden severe MR into a small, stiff LA.	**Acute** = sudden onset.	**Early systole**, decrescendo.	❶ Sudden papillary/chordal rupture or endocarditis ❷ Large volume into normal-size LA → LA pressure rises fast ❸ LV–LA gradient falls late in systole → murmur short/soft ❹ Flash pulmonary edema, cardiogenic shock	↑ **expiration**.	**Post-MI papillary muscle rupture** (inferior MI, posteromedial), endocarditis, chordal rupture.	Apex (may be soft/absent).	New murmur + sudden pulmonary edema after MI → urgent echo.
D · EARLY–DIASTOLIC MURMURS (decrescendo, starts at S2)								
Aortic regurgitation AR	High-pitched, **blowing early-diastolic decrescendo** murmur from backflow **aorta → LV**.	**Aortic** valve · **Regurgitation** = backward flow.	**Early diastole**, starts right at A2, decrescendo.	❶ Leaflet or aortic-root disease → incompetent valve ❷ Aortic diastolic pressure > LV → backflow; gradient largest at start → decrescendo ❸ LV volume overload → eccentric hypertrophy ❹ ↑ stroke volume + ↓ diastolic BP → wide pulse pressure	↑ **expiration**. ↑ handgrip; ↓ amyl nitrite.	Bicuspid, rheumatic, endocarditis, **root dilation** (Marfan, HTN, syphilis, ankylosing spondylitis, dissection).	**Erb's point**/LSB (valve), RSB (root); **sit up, lean forward, end-expiration**, diaphragm.	Bounding pulse: Corrigan, de Musset, Quincke, Duroziez. Acute AR = short murmur.
Pulmonic regurgitation PR · Graham Steell	Early-diastolic decrescendo murmur from backflow **PA → RV**; **Graham Steell** = high-pitched PR from PHT.	**Pulmonic Regurgitation**; Graham Steell = physician who described it (1888).	**Early diastole**, after P2.	❶ PHT → dilated PA & annulus ❷ Incompetent valve → high-pressure backflow ❸ High-pitched murmur (low-pitched if PA pressure normal)	↑ on **inspiration**.	PHT (MS, COPD, PAH), post-TOF repair, endocarditis.	2nd–3rd LICS.	Loud P2 + early diastolic murmur ↑ on inspiration = Graham Steell.
E · MID– / LATE–DIASTOLIC MURMURS (rumbles)								
Mitral stenosis MS	**Low-pitched mid-diastolic rumble** with presystolic accentuation from flow through a narrowed mitral valve.	**Mitral** valve · **Stenosis** = narrowing.	**Mid-diastole after opening snap**; presystolic ↑ (atrial kick) — **lost in AF**.	❶ Rheumatic commissural fusion → ↓ valve area ❷ LA–LV diastolic gradient → turbulent inflow → rumble ❸ ↑ LA pressure → LA enlargement (AF, thrombus, dysphagia, Ortner hoarseness) ❹ Pulmonary venous HTN → PHT → loud P2, RV failure	↑ **expiration**; ↑ after exercise, LLD.	**Rheumatic heart disease** (most), mitral annular calcification; unmasked by pregnancy.	**Apex**, **bell**, **LLD**; very localized.	Loud S1 + OS + rumble. Shorter A2–OS / longer rumble = severe.
Tricuspid stenosis TS	Mid-diastolic rumble through a stenotic tricuspid valve.	**Tricuspid** valve · **Stenosis**.	**Mid-diastole**, presystolic ↑.	❶ Leaflet fusion → RA–RV gradient → rumble ❷ ↑ RA pressure → giant **a waves**, hepatomegaly, edema	↑ on **inspiration**.	Rheumatic (with MS), carcinoid.	**LLSB**, bell.	Giant a waves, slow y descent.
Austin Flint murmur	Apical mid-diastolic rumble in **severe AR** without organic MS.	**Austin Flint** = US physician (1862).	**Mid/late diastole**.	❶ AR jet hits anterior mitral leaflet → partially closes it (functional MS) ❷ ↑ LV diastolic pressure ❸ Turbulent mitral inflow → rumble	↑ **expiration**; ↓ amyl nitrite (MS ↑).	Severe chronic AR.	Apex, bell.	No opening snap, no loud S1 (vs true MS).
Carey Coombs murmur	Soft apical mid-diastolic murmur in **acute rheumatic fever**.	**Carey Coombs** = British physician (1924).	**Mid-diastole**.	❶ Acute rheumatic carditis → mitral valvulitis ❷ Edematous, inflamed leaflets → turbulent inflow → soft rumble	↑ **expiration**.	Acute rheumatic fever (carditis).	Apex, bell.	Transient; no OS; fades with recovery.
Diastolic flow rumbles relative MS / TS	Mid-diastolic rumble from ↑ **flow across a normal AV valve**.	**Relative / functional stenosis** = valve normal, flow too large.	**Mid-diastole**, often after S3.	❶ Large shunt or severe regurgitation ❷ ↑ diastolic volume crosses normal AV valve ❸ Turbulence → short rumble	Mitral flow ↑ expiration; tricuspid flow ↑ inspiration.	**Mitral:** severe MR, VSD, PDA. **Tricuspid:** ASD, severe TR.	Apex (mitral) / LLSB (tricuspid), bell.	Implies a large shunt (Qp:Qs > 2:1).
F · CONTINUOUS MURMURS & RELATED SOUNDS								
Patent ductus arteriosus PDA	**Continuous 'machinery'** murmur from aorta → pulmonary artery through a persistent ductus.	**Patent** = open · **Ductus arteriosus** = fetal vessel linking PA and aorta.	**Continuous** (systole + diastole), peaks around S2.	❶ Ductus fails to close after birth (normally ↑ O_2, ↓ PGE_2) ❷ Aortic > PA pressure all cycle → continuous L→R flow ❸ LA/LV overload; wide pulse pressure ❹ Eisenmenger → **differential cyanosis** (lower limbs)	Little change.	Prematurity, **congenital rubella**, high altitude.	**Left infraclavicular** (1st–2nd LICS).	Indomethacin closes; PGE_1 keeps open.
Venous hum	Benign continuous murmur from turbulent flow in the **jugular veins**.	**Venous** = veins · **Hum** = soft continuous sound.	**Continuous**, louder in diastole.	❶ High-velocity venous return from neck ❷ Turbulence at jugular–subclavian/SVC junction	Louder sitting & inspiration; **gone supine** or with jugular pressure/head turn.	Children, pregnancy, anemia, thyrotoxicosis. (Also **mammary souffle**: late pregnancy/lactation.)	Right supraclavicular.	Only innocent murmur that is continuous.
AV fistula / ruptured sinus of Valsalva	Continuous murmurs from an abnormal **high → low pressure** connection.	**AV** = artery–vein · **Sinus of Valsalva** = aortic root bulge above each cusp.	**Continuous**.	❶ High-pressure vessel connects to low-pressure chamber/vein ❷ Gradient persists in systole & diastole → continuous flow	Little change.	Dialysis fistula, trauma, coronary AV fistula; sinus aneurysm rupture into RA/RV (sudden pain + HF).	Over the lesion; sinus: LSB.	Dialysis fistula: thrill/bruit = patent. Large fistula → high-output HF.
Coarctation of aorta	Systolic (± continuous) murmur from **aortic narrowing**, usually distal to left subclavian.	**Coarctation** (L. coarctare) = to press together.	Systolic (late); continuous from collaterals.	❶ Narrowing → ↑ upper-body BP, ↓ lower-body BP ❷ Turbulent flow at the site → systolic murmur ❸ Intercostal collaterals dilate → continuous murmurs, **rib notching**	Little change.	**Turner syndrome**, bicuspid aortic valve, berry aneurysm.	**Left interscapular** (back).	Radio-femoral delay; arm BP > leg; '3' sign on CXR.
Pericardial friction rub (not a murmur)	Scratchy, grating sound from **inflamed pericardial layers** rubbing.	**Friction rub** = sound of two surfaces rubbing.	Up to **3 components**: atrial systole, ventricular systole, early diastole.	❶ Inflamed visceral & parietal pericardium lose smooth lubrication ❷ Surfaces rub with every cardiac movement	Often ↑ inspiration; **persists with breath-holding** (pleural rub stops).	Acute pericarditis: viral, uremic, post-MI/Dressler, autoimmune, TB, malignancy.	LLSB, **sit up & lean forward**, diaphragm.	Evanescent; may vanish as effusion grows.

Grading (Levine)
Systolic 1–6: **1** faint · **2** soft · **3** loud, no thrill · **4** + **thrill** · **5** heard with edge of stethoscope · **6** heard off chest. Diastolic graded 1–4.

Describe every murmur
Timing (systolic/diastolic/continuous) · **shape** · **location** · **radiation** · **intensity** · **pitch** (bell/diaphragm) · **quality** (harsh, blowing, rumbling) · **maneuvers**.

Radiation cheat sheet
AS → carotids · MR → axilla · PS → left shoulder/back · Coarctation → back · AR → apex/LSB · HOCM & VSD → no carotid radiation.

Red flags → echo
Any **diastolic** or **continuous** murmur (except venous hum) · holosystolic · grade ≥3 or thrill · abnormal S2 · symptoms (syncope, chest pain, dyspnea) · cyanosis.

Mnemonic
Systolic = **"MR. ASS"** (MR, AS) · Diastolic = **"MS. ARD"** (MS, AR). Same logic on the right: TR/PS systolic, TS/PR diastolic.

MASTER KEY OF ALL DISEASES		JUGULAR VENOUS		ARTERIAL				PRECORDIAL				CARDIAC AUSCULTATION				PULMONARY AUSCULTATION
DISEASE STATE	CODE NUMBER	MEAN PRESSURE	WAVE FORM	BLOOD PRESSURE	CAROTIDS	BRACHIALS	FEMORALS	RV	PA	LV	DLV	AORTIC	PULMONARY	TRICUSPID	MITRAL	IN DISEASES WITH PATHOLOGIC FINDINGS, UNDESCRIBED AREAS HAVE NORMAL SOUNDS
Introductory Program	44	3		120/80												Normal vesicular breath sounds
Normal	46	4		120/70												Normal vesicular breath sounds
Innocent Murmur	22	3		110/70												Normal vesicular breath sounds
Aortic Valve Sclerosis	49	3		165/80												Normal vesicular breath sounds
Hypertension	36	5		170/110												Normal vesicular breath sounds
Angina Pectoris	39	4		150/100												Normal vesicular breath sounds
Acute Inferior Myocardial Infarction	43	3		90/60												Lower posterior bilateral inspiratory crackles
Acute Anterior Myocardial Infarction	40	6		120/95												Inspiratory and expiratory crackles
Ventricular Aneurysm	48	7		150/90												Normal vesicular breath sounds
Mitral Valve Prolapse	9	3		120/70												Normal vesicular breath sounds
Mitral Valve Prolapse, isolated click & murmur	10	3		120/70												Normal vesicular breath sounds
Mitral Regurgitation, chronic	7	7		130/80												Normal vesicular breath sounds
Mitral Regurgitation, mild	8	4		120/70												Inspiratory and expiratory crackles
Mitral Regurgitation, acute	6	5		140/80												Inspiratory crackles and expiratory wheezes
Mitral Stenosis with severe tricuspid regurgitation	4	12		90/70												Inspiratory crackles and expiratory wheezes
Mitral Stenosis with mild tricuspid regurgitation	5	12		90/70												Lower bilateral inspiratory crackles and expiratory wheezes
Mitral Stenosis and Regurgitation	3	11		120/80												Normal vesicular breath sounds
Aortic Regurgitation, chronic	17	7		160/35												Lower bilateral inspiratory and expiratory crackles
Aortic Regurgitation, acute	37	6		130/50												Normal vesicular breath sounds
Aortic Stenosis	13	3		100/80												Normal vesicular breath sounds
Hypertrophic Obstructive Cardiomyopathy	1	5		120/80												Lower right posterior absent breath sounds, and lower left posterior inspiratory crackles
Cardiomyopathy	42	12		110/90												Lower left pleural rub
Acute Pericarditis	30	5		130/85												Normal vesicular breath sounds
Primary Pulmonary Hypertension	11	12		90/70												Normal vesicular breath sounds
Atrial Septal Defect	23	6		120/80												Normal vesicular breath sounds
Ventricular Septal Defect	25	3		100/60												Normal vesicular breath sounds
Patent Ductus Arteriosus	28	7		150/50												Normal vesicular breath sounds
Pulmonary Stenosis	20	10		110/80												Normal vesicular breath sounds
Coarctation of the Aorta	33	4		170/100												Normal vesicular breath sounds
Tetralogy of Fallot	27	6		90/60												Normal vesicular breath sounds